Robots for Those in Transition: Exploring Mental Wellness through Design Probes

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BACKGROUND

Emerging adulthood brings unique challenges and significant life transitions.
These transitional challenges affect students' mental wellness. [1]

RELATED WORK

Current HRI and HCI research focuses on exploring using social robots to support mental wellness in accessible ways. [2, 3, 4]

Design probe is an important method in HRI & HCI for involving vulnerable populations about sensitive topics. [5, 6, 7]

RESEARCH GOALS

RQ1. What design outcomes do international students generate when using a co-design probe kit carefully tailored to their context?

RQ2. What are the implications for a mental wellness robot that can be integrated into international students' daily lives?

METHODS

We deployed take-home co-design probes with 18 international students to investigate their lived experiences and coping strategies and translate these insights into design implications for social robots.

Participants completed individual design activities over two weeks; photographs of their artifacts and interview transcripts were analyzed using thematic analysis.



The co-design probe kit

DESIGN IMPLICATIONS

Seamless integration into daily routines. Low-effort activation and engagement that allow seamless integration into users' daily lives.

Context-specific guidance. More realistic, context-specific guides that could reduce anxiety about integrating into a new culture, presented as familiar languages, accessible formats, and tactile experiences.

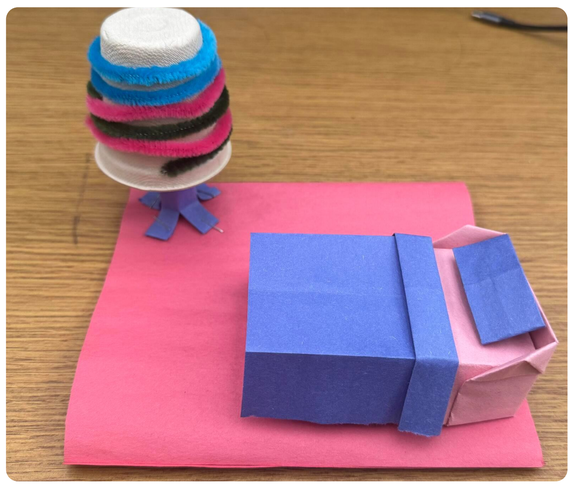
Agency through controllable tasks. Being organized, working on small tasks, caring for others, and engaging in simple activities to ease their nerves.

External features as a core design element. Soft, fluffy textures, bright colors, and welcoming gestures, such as smiling faces and hugging gestures.

RESULTS

Design probe outcomes created and shared by participants.

Robots as Part of the Room



Warm light and sleep (P17)

Creating a welcoming living environment through seamless design.

Robots for Learning Social Norms



Cultural understanding (P11)

Accessible guidance about social norms in non-academic, daily contexts.

Robots for Agency through Controllable Tasks



Caring for birds (P5)

Regain a sense of agency through small practices and caring for others.

Robots Providing Companionship



Bright colors and a welcoming appearance (P4)

Companionship without requiring relationship-building efforts.

DISCUSSION

International students prefer easy-to-use products that integrate seamlessly into daily life, envisioning robots that fit naturally into existing environments and routines.

Our approach is better understood as probe-based exploratory research than as collaborative co-design.

Future study: Incorporate collaborative sessions while minimizing participants' stigma and discomfort when disclosing sensitive information.

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